

# Basin MCMC: Discussion

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*This document is meant to serve as a discussion of the Basin MCMC algorithm.*

## 1 What kind of algorithm do we want?

In general, we want an algorithm that correctly samples from a target density  $\pi(\cdot)$  (i.e., produces a Markov chain that is ergodic with respect to  $\pi(\cdot)$ ). Moreover, we would like the chain to converge to the target density within a reasonable number of iterations. Moreover, out of all such algorithms, we would like to minimize the computational cost per effective sample.

The standard algorithm that indeed correctly samples from a target density  $\pi(\cdot)$  is the Metropolis–Hastings (MH) algorithm.

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**Algorithm 1** Metropolis–Hastings (MH)

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**function:**  $[x^1, \dots, x^N] = \text{MH}(\pi(\cdot), q(\cdot | \cdot), x^0, N)$

**input:** target density  $\pi(\cdot)$ , proposal density  $q(\cdot | \cdot)$ , initial state  $x^0$ , number of steps  $N$

**output:** ordered list of states  $[x^1, \dots, x^N]$

- 1: **for**  $j = 0$  to  $N - 1$  **do**
- 2:     Given  $x^j$ , generate a proposal  $y$  distributed as  $q(y | x^j)$ .
- 3:     Accept proposal  $y$  as the next state, i.e. set  $x^{j+1} = y$ , with probability

$$\alpha(y | x^j) = \min \left\{ 1, \frac{\pi(y)q(x^j | y)}{\pi(x^j)q(y | x^j)} \right\}.$$

- 4:     Otherwise reject  $y$  and set  $x^{j+1} = x^j$ .
  - 5: **end for**
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However, when the target density  $\pi(\cdot)$  is multimodal (e.g., as is the case when we use high-level representations for states), with basic choices of proposal kernel  $q(\cdot | \cdot)$ , the MH algorithm can struggle to explore all modes within a reasonable number of iterations. In this case, the MH algorithm will not converge to the target density within a reasonable number of iterations.

One way to address this is to represent the target density as a collection of local probability distributions, which we call *basins*. Each basin has a representative state and describes states that are similar to that representative state. The idea is that each basin captures one local region, or mode, of the target density.

Instead of directly moving between individual states using only basic proposals (e.g., birth, death, and perturbation moves), we treat the state of the algorithm as a collection of basins. We then perform moves that add, remove, perturb, merge, or split these basins. The perturb move runs a local sampler within a basin that explores

nearby states and adapts the basin’s probability distribution and probability so that it better represents the local shape of the target density.

I loosely view this algorithm as fitting the proposal kernel to the target density, including all its modes. Local moves allow the algorithm to fit a discovered mode, while birth, death, merge, and split moves allow it to discover new modes and change its representation of the existing modes. Moreover, since this algorithm represents the proposal kernel as a collection of local probability distributions (that we shall choose to be unimodal and easy to sample from), we can sample independently and efficiently from the proposal kernel.

The hope is that this allows the algorithm to explore a multimodal target density more effectively than directly applying basic proposal kernels to individual states. That is, to allow the chain to converge to the multimodal target density within a reasonable number of iterations. And, since the proposal kernel will be close to the target density and give samples that are effectively independent of each other, samples are effectively independent, and so the computational cost per effective sample is reduced.

Once this is achieved, one can worry about tweaking the algorithm to further reduce the computational cost per effective sample by decreasing the cost per sample (e.g., by using a delayed acceptance step to reduce the number of times we need to evaluate the forward map).

This idea is implemented in the Basin MCMC algorithm.

## 2 The Basin MCMC algorithm

### 2.1 Basins

We first discuss the notion of a basin. One way to think of a basin is as a probability distribution over states that captures how *similar* states are to some representative state. In particular, states are considered to be similar to the representative state (e.g., they have high probability density within its basin) if they (are similar in spatial location and) produce similar data (say, if we treat the data generated by the representative state as true data, then the posterior probability density of the state given the data is high).

To be more precise, a basin  $B$  is a set of the following:

- A **representative state**  $x_0$ .
- A **probability distribution**  $\pi_B(\cdot)$  over states. We can think of this as consisting of:
  - A **sequence of probability distributions**  $\{\pi_{B,n}(\cdot)\}_{n \in \mathbb{N}}$  each over the set of states with  $n$  atoms  $\mathcal{X}_n$ . Note that in cases where the order of states does not matter (as it does not in our GPR application), each  $\pi_{B,n}(\cdot)$  can be entirely characterised as a distribution over each component of an atom.
  - A **sequence of probabilities**  $\{p_{B,n}\}_{n \in \mathbb{N}}$  each defining the probability of a state having  $n$  atoms, given the state is in basin  $B$  (e.g., it is distributed by  $\pi_B(\cdot)$ ).
- A **probability**  $p_B$  that the state is in basin  $B$  (e.g., it is distributed by  $\pi_B(\cdot)$ ).

- (Maybe?) a **local sampler** that generates samples from the posterior, proposed by  $\pi_B(\cdot)$ , and adapts  $\pi_B(\cdot)$  and  $p_B$  with each iteration.

## 2.2 Moves on basins

Recall that our original MCMC algorithm worked by doing a mixture of moves between states of the model, such as birth, death, and perturbation moves. The basin MCMC algorithm will work in a similar way, still doing some kind of mixture of moves, but instead we will consider the state of the MCMC to be the set of basins (which implies a state of the model by unioning the representative states of the basins), and we will do moves on these basins.

The moves we will do on basins are:

- **Birth move:** add a new basin to the state.
- **Death move:** remove a basin from the state.
- **Perturb move:** run the local sampler within the basin, adapting the probability distribution and probability of the basin, and update the representative state of the basin to be the final state of the local sampler.
- **Merge move:** merge two basins into one, where the representative state of the new basin is the union of the representative states of the two basins.
- **Split move:** split a basin into two, where the representative state of the new basins is a partition of the representative state of the original basin.

To ensure the mixture of moves is in detailed balance with the target density it suffices to ensure each of the moves is in detailed balance with the target density. To achieve this, we must combine the birth and death moves, and the merge and split moves, into a single move. We then ensure each of the birth/death, perturb, and merge/split moves is in detailed balance with the target density with an MH step.

## 2.3 Algorithm

The algorithm roughly works as follows:

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**Algorithm 2** Basin MCMC

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**function:**  $[x^1, \dots, x^N] = \text{BasinMCMC}(\pi(\cdot), \{\pi_{C,x}(\cdot)\}_{x \in \mathcal{X}}, q(\cdot | \cdot), n, x^0, N)$   
**input:** target density  $\pi(\cdot)$ , birth/death proposal kernel  $q_{\text{Birth/Death}}(\cdot | \cdot)$ , perturb proposal kernel  $q_{\text{Perturb}}(\cdot | \cdot)$ , merge/split proposal kernel  $q_{\text{Merge/Split}}(\cdot | \cdot)$ , initial state  $x^0$ , initial basin probability distribution  $\pi_B(\cdot)$  (dependent on representative state  $x^0$ ), number of steps  $N$   
**output:** ordered list of states  $[x^1, \dots, x^N]$

- 1: Initialize list of basins  $\mathcal{B}^0 = [B]$  with  $B = (x^0, \pi_B(\cdot), p_B = 1)$ .
- 2: **for**  $j = 0$  to  $N - 1$  **do**
- 3:     Given  $\mathcal{B}^j$ , choose a move from  $\{\text{Birth/Death}, \text{Perturb}, \text{Merge/Split}\}$ .
- 4:     **if** Birth/Death: **then**
- 5:         Propose a list of basins  $\mathcal{B}^*$  from  $q_{\text{Birth/Death}}(\cdot | \mathcal{B}^j)$ , where  $\mathcal{B}^*$  is  $\mathcal{B}^j$  with an additional or removed basin.
- 6:         Compute MH acceptance probability  $\alpha(\mathcal{B}^* | \mathcal{B}^j)$  for Birth/Death move.
- 7:     **else if** Perturb: **then**
- 8:         Propose a list of basins  $\mathcal{B}^*$  from  $q_{\text{Perturb}}(\cdot | \mathcal{B}^j)$ , where  $\mathcal{B}^*$  is  $\mathcal{B}^j$  with some basin's probability distribution and probability adapted, and representative state updated. This is done by running the local sampler of some basin.
- 9:         Compute MH acceptance probability  $\alpha(\mathcal{B}^* | \mathcal{B}^j)$  for Perturb move.
- 10:     **else if** Merge/Split: **then**
- 11:         Propose a list of basins  $\mathcal{B}^*$  from  $q_{\text{Merge/Split}}(\cdot | \mathcal{B}^j)$ , where  $\mathcal{B}^*$  is  $\mathcal{B}^j$  with two merged or split basins.
- 12:         Compute MH acceptance probability  $\alpha(\mathcal{B}^* | \mathcal{B}^j)$  for Merge/Split move.
- 13:     **end if**
- 14:     Accept proposal  $\mathcal{B}^*$ , i.e. set  $\mathcal{B}^{j+1} = \mathcal{B}^*$ , with probability  $\alpha(\mathcal{B}^* | \mathcal{B}^j)$ .
- 15:     Otherwise reject  $\mathcal{B}^*$  and set  $\mathcal{B}^{j+1} = \mathcal{B}^j$ .
- 16:     Set  $x^{j+1} = \bigcup_{B \in \mathcal{B}^{j+1}} B$ .
- 17: **end for**

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## 2.4 Development

This can become a fairly sophisticated algorithm. To develop it, I will complete the following tasks, implementing them in Python as I go:

1. **Decide initial basin probability distribution and probability:** Decide how to choose the initial probability distribution and probability for a basin given an initial representative state.
2. **Decide birth/death move:** Decide what the birth/death move should be.
3. **Decide perturb move:** Decide what the perturb moves should be. That is, decide what the local sampler for each basin should be. This includes how we propose from the basin probability distribution, and how we adapt the basin probability distribution and probability.
4. **Decide merge/split move:** Decide what the merge/split move should be.
5. **Decide move choice:** Decide how to choose a move from the set of possible moves.
6. **Prove diminishing adaption:** Prove that the adaption is diminishing, and so the chain is ergodic. Note that by construction, this MCMC is ergodic without

any adaption (as long as the birth move shares support with the target density, etc).

If I have extra time, I will also complete the following tasks:

1. **Include adaption stopping mechanism:** Decide a mechanism for stopping adaption of basins once they have diminishing adaption.
2. **Include delayed acceptance step:** Decide a delayed acceptance step before computing the MH acceptance probability and doing the MH step. We would use a bogus likelihood for the delayed acceptance step, and then use the actual likelihood for the MH step. This would reduce the number of times we need to evaluate the forward map and should increase efficiency with a well chosen bogus likelihood.
3. **Consider adaption of probabilities:** Consider adding a step after acceptance/rejection that given  $\alpha(\mathcal{B}^* | \mathcal{B}^j)$ , adapts the probabilities of the basins in  $\mathcal{B}^{j+1}$ . I will probably not do this as it complicates the algorithm. I am only considering this as one may want to use all evaluations of the forward map (as is done to evaluate  $\alpha(\mathcal{B}^* | \mathcal{B}^j)$ ) to adapt the basins.